BumbleBee
INSTRUMENTS PRIVATE LIMITED
An IIT Palakkad Incubated Company

SCOPE - X

Electronics at Home

Technical Manual

Complete Technical Manual

• Bipolar Supply $\pm$ 12V added • Offset and Gain adjust simplified

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1 Introduction

Overview: The Scope-X development board from Bumblebee Instruments provides a comprehensive platform for understanding basic analog circuit design principles through hands-on experimentation and analysis.

This manual serves as both an instructional guide and reference document for the Scope-X. Designed by Bumblebee Instruments, the kit provides students and engineers with practical experience in analog circuit design, analysis, and implementation using industry-standard components and methodologies.

The laboratory exercises contained within this manual progress from fundamental operational amplifier concepts to advanced signal processing techniques, ensuring a thorough understanding of analog system design principles.

1.1 Learning Objectives

Upon completion of this manual, users will be able to:

- Design and analyze operational amplifier circuits
- Implement active filter topologies
- Understand analog signal processing fundamental
- Troubleshoot analog circuit implementations

1.2 Kit Components

Table 1: Hardware Components Overview

Component	Description	Quantity
Scope-X Main Board	Analog signal acquisition and processing unit	1
USB Interface Cable	Enables communication with the PC	1

2 System Setup and Configuration

Important: Ensure all power connections are verified before energizing the system. Observe proper handling procedures when working with electronic components.

2.1 Setup Procedure

Prerequisites:

- A computer with a valid USB port.
- The Scope-X Software Suite and Drivers (available for download).
- Administrator access for driver installation.

Quick Start Guide

1. **Software Installation:** Install the Scope-X application and the requisite USB drivers for your specific operating system (Windows, Linux, or macOS) from the provided resource folder.
2. **Hardware Connection:** Connect the Scope-X board to your computer using the USB cable.
3. **Device Recognition:** Allow your operating system to initialize the device. It should appear as a USB Serial/COM device.
4. **Launch:** Open the Scope-X GUI. Use the 'Check Ports' or 'Connect' feature to establish communication with the board.

Connection Status

When successfully connected, the application will display the Port Number to which the device is connected and a Disconnect option.

3 GUI Software Reference

The Scope-X GUI provides a centralized interface for signal generation, acquisition, and analysis. (QT version). A detailed Application Manual provided a detailed description is available. Click here

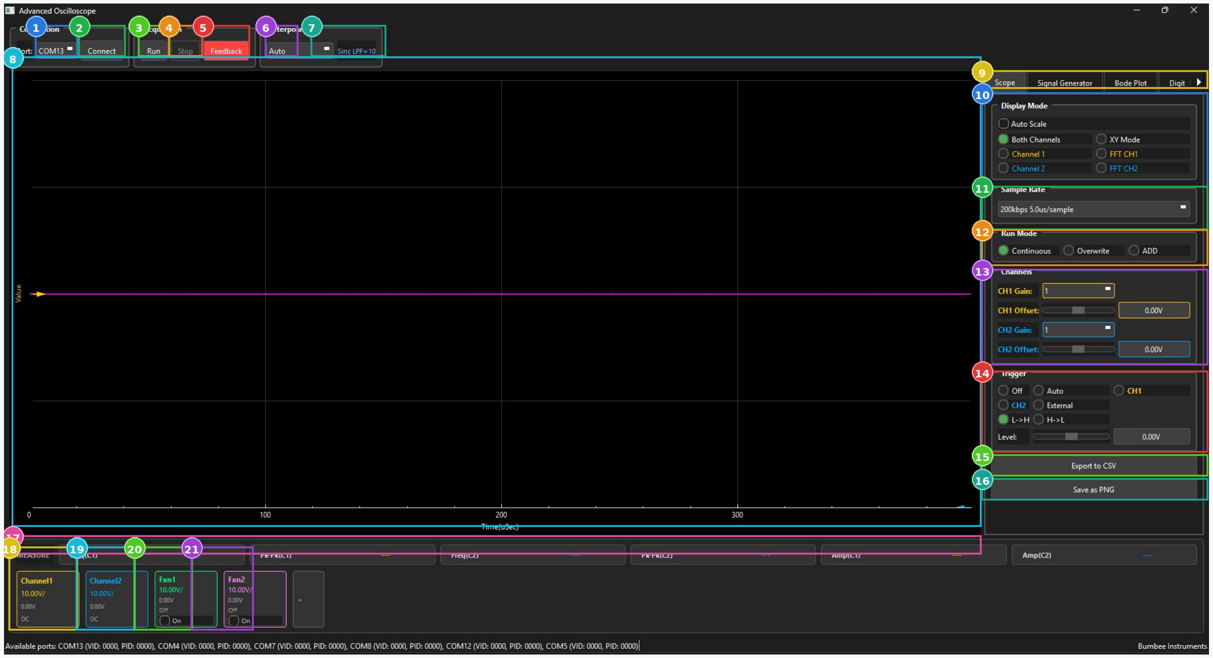

Figure 1: Software Interface

4 Experiment 1: Inverting Amplifier

Objective: Design an amplifier with a Gain of -10 using IC 741.

Theory and Design

The inverting amplifier provides a phase-shifted output (180°) amplified by the ratio of the feedback resistor to the input resistor.

Formula:

$$V_{out} = -\left(\frac{R_f}{R_{in}}\right) V_{in} \tag{1}$$

Design Parameters:

- **Target Gain (A_v):** -10
- **Let R_{in}:** 1 kΩ
- **Calculate R_f:**

$$R_f = |A_v| \times R_{in} = 10 \times 1\text{k} = 10\ \text{k}\Omega$$

Procedure:

1. Connect Op-Amp Pin 2 to R_{in}.
2. Connect R_f between Pin 2 and Pin 6.
3. Ground Pin 3 (Non-Inverting Input).

Circuit Diagram

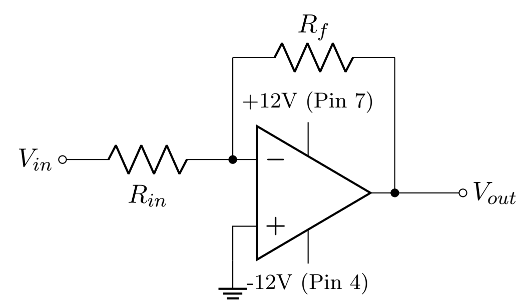

Figure 2: Inverting Amplifier Connection

5 Experiment 2: Non-Inverting Amplifier

Objective: Design an amplifier with a Gain of 5 using IC 741.

Theory and Design

The non-inverting amplifier provides amplification without phase inversion. The input signal is applied directly to the non-inverting terminal.

Formula:

$$V_{out} = \left(1 + \frac{R_f}{R_1}\right) V_{in} \tag{2}$$

Design Parameters:

- **Target Gain (A_v):** 5
- **Let R_1:** 1 kΩ
- **Calculate R_f:**

$$5 = 1 + \frac{R_f}{1k} \implies \frac{R_f}{1k} = 4$$

$$R_f = 4 \text{ k}\Omega$$

Procedure:

1. Apply V_{in} to Pin 3.
2. Connect voltage divider (R_1, R_f) to Pin 2.
3. Ground the other end of R_1.

Circuit Diagram

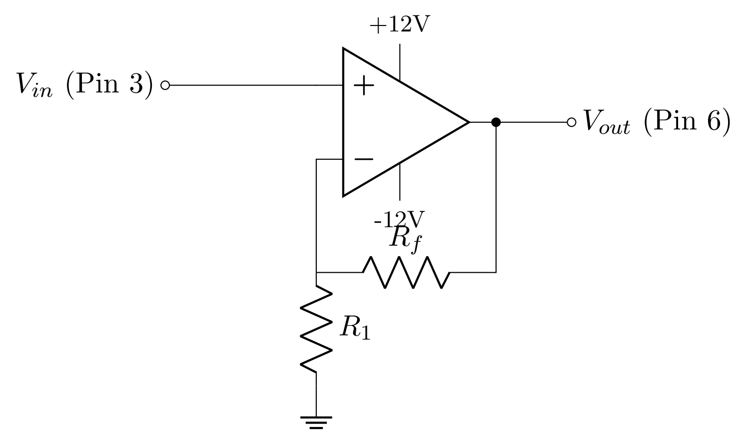

Figure 3: Non-Inverting Amplifier Connection

6 Experiment 3: Active Low-Pass Filter

Objective: Design a filter to attenuate frequencies above 1 kHz.

Theory and Design

An active Low-Pass Filter (LPF) allows low frequency signals to pass but blocks high frequencies. By placing a capacitor in parallel with the feedback resistor, the gain decreases as frequency increases.

Formula (Cut-off Frequency):

$$f_c = \frac{1}{2\pi R_f C_f} \tag{3}$$

Design Parameters:

- **Target f_c:** ≈ 1 kHz
- **Let $R_f = R_{in}$:** 10 kΩ (Unity DC Gain)
- **Calculate C_f:**

$$C_f = \frac{1}{2\pi(10k)(1k)} \approx 15.9 \text{ nF}$$

Use standard value: 15 nF or 16 nF.

Circuit Diagram

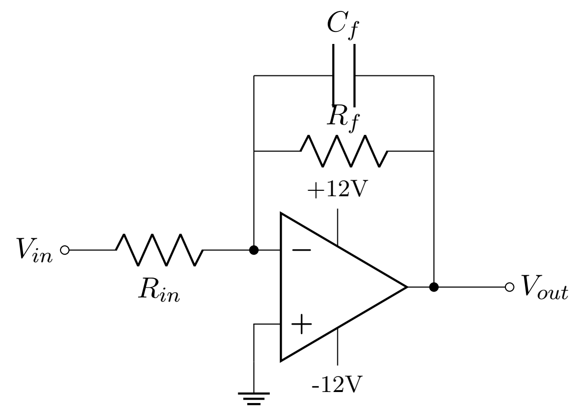

Figure 4: Active LPF (Inverting Topology)

7 Experiment 4: Active High-Pass Filter

Objective: Design a filter to block DC and low frequencies (Cut-off 100 Hz).

Theory and Design

An active High-Pass Filter (HPF) blocks low frequencies. This is achieved by placing a capacitor in series with the input resistor. At low frequencies, the capacitor impedance is high, blocking the signal.

Formula:

$$f_c = \frac{1}{2\pi R_{in} C_{in}} \tag{4}$$

Design Parameters:

- **Target f_c:** $\approx$ 100 Hz
- **Let R_{in}:** 10 kΩ
- **Calculate C_{in}:**

$$C_{in} = \frac{1}{2\pi(10k)(100)} \approx 159 \text{ nF}$$

Use standard value: 160 nF or 0.16μF.

Circuit Diagram

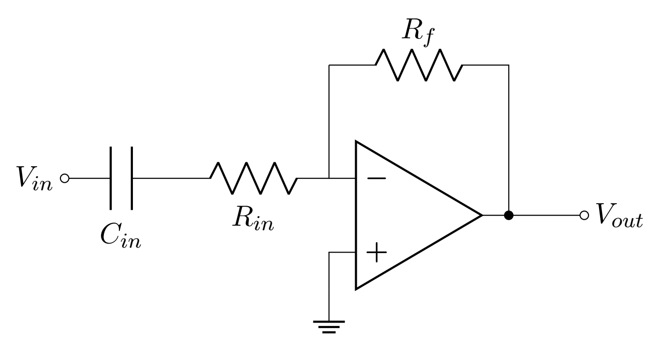

Figure 5: Active HPF (Inverting Topology)

Troubleshooting Guide

Symptom	Solution
Clipping Output	Input signal is too large. Reduce input amplitude or gain.
No Signal (Flatline)	Check power rails ($\pm$12V) and breadboard connectivity.
50Hz/60Hz Noise	Ensure the board Ground (GND) is connected to the circuit ground.